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رقم الكمبيوتر * □□□□

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الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) □□

KHWH

Will be reviewed

8

0.5-point

Which statement is correct about Distance Vector routing? □□

- ☐ Nodes have to calculate the costs to only direct neighbouring nodes
☒ Nodes have to calculate the costs to all the nodes in the network
☐ None of the above

Will be reviewed

9

1.5-point

Question □□

Suppose that there are M paths between the server and the client. No two paths share a link. Path k ($k = 1, \dots, M$) consists of N links with transmission rates, $R_1^k, R_2^k, \dots, R_N^k$. If the server can use all M paths to send data, the maximum throughput that the server can achieve is.....

- a. $\min\{R_1^k, R_2^k, \dots, R_N^k\}$
 b. $\max\{R_1^k, R_2^k, \dots, R_N^k\}$
 c. $\sum_{k=1}^M \min\{R_1^k, R_2^k, \dots, R_N^k\}$
 d. $\max\{\min\{R_1^1, R_2^1, \dots, R_N^1\}, \min\{R_1^2, R_2^2, \dots, R_N^2\}, \dots, \min\{R_1^M, R_2^M, \dots, R_N^M\}\}$
 e. $\sum_{k=1}^M \max\{R_1^k, R_2^k, \dots, R_N^k\}$
 f. $\min\{\max\{R_1^1, R_2^1, \dots, R_N^1\}, \max\{R_1^2, R_2^2, \dots, R_N^2\}, \dots, \max\{R_1^M, R_2^M, \dots, R_N^M\}\}$
 g. None of the above

- ☐ a

- ☐ b
- ☒ c
- ☐ d
- ☐ e
- ☒ f
- ☐ g

Will be reviewed

10 0.5-point

To perform load distribution, when clients make a DNS query for a name mapped to a set of addresses, the server responds with the entire set of IP addresses, but rotates the ordering of the addresses within each reply. ☐

- ☒ True
- ☐ False

Will be reviewed

11 1-point

Suppose that the TCP receiver computes the internet checksum for the received TCP segment and finds that it matches the value carried in the checksum field. Accordingly, the receiver should be certain that no bit errors have occurred. ☐

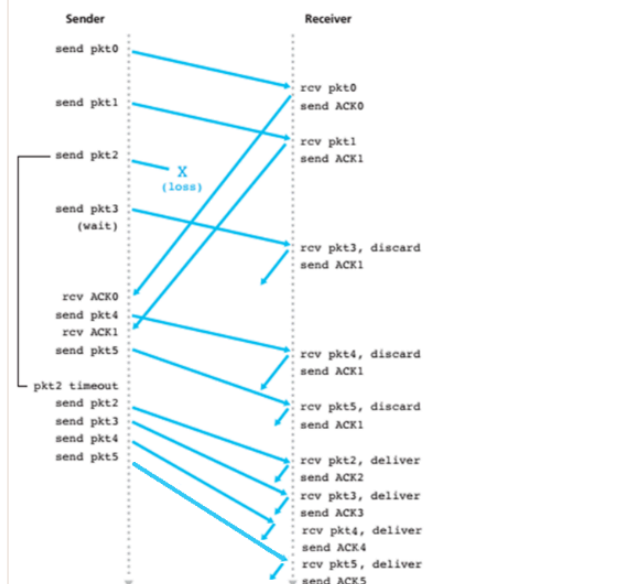
- ☐ True
- ☒ False

Will be reviewed

12

Question ☐

The following figure represents correct operation of the GBN protocol for the case of a window size of 4 packets. Consider that there is no problem that the protocol starts the operation by sending from pkt0.



- ☒ True
- ☐ False

2-point

Will be reviewed

13 0.5-point

A switch is a plug and play device builds its table automatically in which for each incoming frame received on an interface ,the switch stores the following three items in its table: (1) MAC address in the frame destination address field (2) the interface from which the frame arrived (3) current time ☐

☒ True

☐ False

Will be reviewed

14 2-point

Consider that ISP has been allocated the address block 200.23.16.0/23 and wants to divide this block into 4 equal sized contiguous address blocks. Then the following are correct blocks addresses: 200.23.16.0/25 ; 200.23.16.128/25 ; 200.23.17.0/25; 200.23.17.128/25. ☐

☒ True

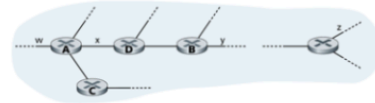
☐ False

Will be reviewed

15

Question ☐

Suppose the following network and consider the RIP protocol.



Suppose the following figure shows the routing table for router "D"

Destination Subnet	Next Router	Number of Hops to Destination
W	A	2
Y	B	2
Z	B	4
X	---	1
----	----	----

If router "D" receives from router "A" the advertisement shown in the following figure

Destination Subnet	Next Router	Number of Hops to Destination
Z	C	4
W	---	1
X	---	1
----	----	----

Then router "D" will update its routing table, as shown in the following figure

Destination Subnet	Next Router	Number of Hops to Destination
W	A	2
Y	B	2
Z	A	5
X	---	1
----	----	----

☐ True

☒ False

2-point

Will be reviewed

16 1-point

In circuit switching, link transmission is shared on packet-by-packet basis among only users who have packets that need to be transmitted over the link. ☐

☐ True

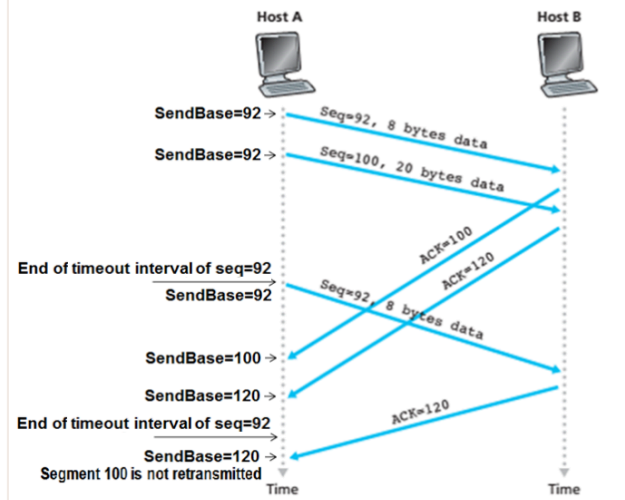
☒ False

Will be reviewed

17

Question ☐

The following figure represents correct operation of the TCP protocol.



☐ True

2.5-point

☒ False

Will be reviewed

18 1-point

If IPv4 datagram of 5000 bytes is fragmented to fit MTU of 1500 bytes, then [?]

- ☐ there will be 3 fragments with offsets 0, 185, 370 respectively
- ☐ there will be 4 fragments with offsets 0, 1480, 2960, 4440 respectively
- ☐ there will be 4 fragments with offsets 0, 185, 370, 555 respectively
- ☐ there will be 3 fragments with flag 1, 1, 0, respectively
- ☒ there will be 4 fragments with flag 0, 0, 0, 1 respectively
- ☐ None of the above

Will be reviewed

19 1-point

Consider that there are two hosts, "A" and "B", separated by router, S, and suppose that the ARP table of the host "A" is empty. In this case, if host "A" needs to send frame to host "B", it acquires the MAC address for the interface of the host "B" and sends its IP datagram encapsulated in frame whose destination MAC is that of the node "B" that responds to the ARP query. [?]

☒ True

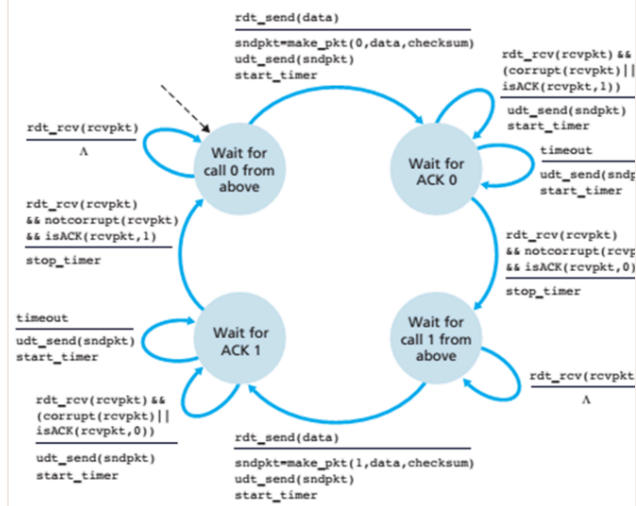
☐ False

Will be reviewed

20

Question [?]

The following FSM for rdt3 sender side is working in a correct and good way to premature (early) timeout.



☐ True

3-point

☒ False

Will be reviewed

21 1-point

In NAT, router must replace Source IP address and Source Port # of every outgoing datagram to NAT IP address and New Port #. Also, it replaces NAT IP address and New Port # in the destination fields of every incoming datagram with the corresponding Source IP address and Port # stored in NAT table. ☐

☒ True

☐ False

Will be reviewed

22 1-point

Question ☐

Consider the following forwarding table in the router

Prefix Match	Link Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
otherwise	3

Then the destination addresses: 11001000 00010111 00011000 10101010 will be forward

☐ Link Interface 0

☒ Link Interface 1

☐ Link Interface 2

☐ Link Interface 3

☐ None of the above

Will be reviewed

23 0.5-point

Transport layer segment includes source port number to be used for de-multiplexing in case of TCP and UDP. ☐

☐ True

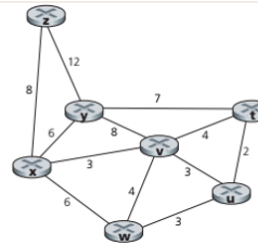
☒ False

Will be reviewed

24

Question ☐

Consider the following network with the indicated link costs. Then using Dijkstra's shortest-path algorithm to compute the shortest path from node "z" to all network nodes will result in the indicated values in the following table.



Step	N'	$D(x), p(x)$	$D(u), p(u)$	$D(v), p(v)$	$D(w), p(w)$	$D(y), p(y)$
0	z	8,z	∞	∞	∞	12,z
1	ZX		∞	11,x	14,x	12,z
2	ZXV		14,v		14,x	12,z
3	ZXVY		14,v		14,x	
4	ZXVYU				14,x	
5	ZXVYUW					
6	ZXVYUWT					

☒ True 3-point

☐ False

☒ Will be reviewed

25 0.5-point

DHCP discover, DHCP offer, and DHCP request messages between client and servers on the same subnet are all broadcasted and sent using UDP protocol. But DHCP ACK message is sent unicast to the client. ☐

☒ True

☐ False

☒ Will be reviewed

26 2.5-point

Consider a packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is 75% done being transmitted on this outbound link and 5 other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1200 bytes and the link rate is 2Mbps. Then the queuing delay of a packet will be approximately 25.2 ms. ☐

☒ True

☐ False

☒ Will be reviewed

27 1-point

UDP has a fine application-level control over when data is sent because under UDP as soon as an application process passes data to UDP, UDP will package the data inside UDP segment and immediately pass the segment to the network layer. ☐

☐ True

☒ False

☒ Will be reviewed

28 1-point

Suppose that a Web server runs in Host "C" on port 80. Also, assume that this Web server uses persistent connections, and is currently receiving requests carry by luck the same source port number from two different Hosts, "A" and "B". Then all of the requests will be sent through the same socket at Host "C". ☐

☐ True

☒ False

☒ Will be reviewed

29 0.5-point

Throughout a session, the FTP server must maintain state about the user. In particular, the server must associate the control connection with a specific user account, and the server must keep track of the user's current directory of the user. But HTTP, on the other hand, is stateless because it does not keep track of any user state. ☐

☒ True

☐ False

Will be reviewed

30 1-point

If the proxy server does not have the object that is requested by the browser, it directly opens a TCP connection with the server name that is mentioned in the "Host" header in the HTTP request message that is received from the browser. But if it has the object, it sends it directly to the browser. ☐

☐ True

☒ False

Will be reviewed

31 0.5-point

TDMA has the following characteristic if M nodes share a link of rate R bps. ☐

☐ When only one node has data to send, it can send at rate R bps.

☒ When M nodes have data to send, each can send with a rate of R/M

☐ It allows some collisions at low load

☒ It uses master node that selects each of the nodes in a round-robin fashion

☐ None of above

Will be reviewed

32 1-point

Unlike POP3, IMAP protocol provides commands to allow users to create folders and move messages from one folder to another. Also, unlike POP3, an IMAP server maintains state information across IMAP sessions. ☐

☒ True

☐ False

Will be reviewed

33 0.5-point

Suppose the client decides to close the TCP connection. Accordingly, the client TCP sends a special TCP segment that has a FIN bit set to 1 to the server process. Then, when the server receives this segment, it sends immediately to the client its own shut-down segment, which has the FIN bit set to 1. ☐

☐ True

☒ False

Will be reviewed

34 2-point

Consider the unusual case in which HTTP is configured to use back-to-back UDP instead of using TCP. Assume that you click on a link in your web browser to obtain a base HTML object from a server. This HTML file references 3 objects, each is located on that server. Let RTT be the round-trip time between your host and the server. By ignoring the transmission delay, then the time it takes from when you click the link until you receive the complete webpage will be ☐

☐ RTT

☒ 2RTT

☐ 3RTT

☐ 4RTT

☐ 5RTT

☐ 6RTT

☐ 8RTT

☐ None of the above

Will be reviewed

35 0.5-point

The count-to-infinity problem in Distance Vector (DV) algorithm under certain conditions may be solved by 

- ☐ randomizing the time in which the nodes send out the distance vectors
- ☐ ensuring that all routers do not run DV algorithm at the same time
- ☒ adding Poisoned Reverse
- ☐ None of the above

Will be reviewed

36 1-point

Traceroute program always gives delay to the first router less than the delay to the second one along the route. 

- ☐ True
- ☒ False

Will be reviewed

37 0.5-point

Internet can be defined as a communication infrastructure provides services to distributed applications and it consists of 4 components: 1) End systems 2) Media 3)Core 4)Protocols. 

- ☒ True
- ☐ False

Will be reviewed

38 0.5-point

Suppose host B's receive buffer becomes full, so that its RcvWindow = 0. After advertising RcvWindow = 0 to host "A", suppose that host "B" has nothing to send to host "A". Accordingly, host "A" is blocked based on TCP specification and doesn't transmit any data of any size to host B. 

- ☐ True
- ☒ False

Will be reviewed

39 1-point

Unlike HTTP, SMTP requires each message, including the body of each message, to be in 7-bit ASCII format. But similar to HTTP, SMTP sends its control information in-band. 

- ☒ True
- ☐ False

Will be reviewed

40 1-point

Suppose a frame with a destination address 49-BD-D2-C7-56-2A arrives at the switch on interface "X". If there is an entry in the table associating 49-BD-D2-C7-56-2A with interface "X", then the switch performs its forwarding function by putting the frame in the output buffer that precedes interface "X". 

- ☒ True
- ☐ False

Will be reviewed

41 1-point

In switching via memory, an input port with an arriving packet first signals the routing processor via an interrupt. The packet is then copied from input port into processor memory. The routing processor then extracts the destination address from the header, looks up for the

appropriate output port in the forwarding table, and copy the packet to the output port buffer. ☐

☒ True

☐ False

☒ Will be reviewed

42 2.5-point

Question ☐

Consider a packet of length L which begins at end system “A” and travels over 4 links to destination end system “B”. These 4 links are connected by three packet switches. Suppose two of the packet switches use store-and-forward while the other one uses cut-through. Let s_i , and R_i denote the length, propagation speed, and the transmission rate of link i , for $i = 1, 2, 3, 4$ respectively. Ignore the processing delay and suppose that $R_1 = R_2 = R_3 = R_4 = R$. The total end-to-end delay for the packet will be $\frac{4L}{R} + \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + \frac{d_4}{s_4}$.

☐ True

☒ False

☒ Will be reviewed

43 3-point

Consider a sender that can transmit at a rate of 1Mbps in both directions. Suppose packets containing only control (e.g., ACK or 3-way handshaking or request) are 100 bits long. Consider the HTTP protocol is used to download a webpage. Suppose that the initial downloaded HTML object contains 3 referenced objects from the same server. Suppose that each downloaded object is 100 Kbits long except the last one which has 200Kbits. Let RTT denotes the round trip time between the local host and the server containing the objects. Then the total time needed to download the objects assuming parallel non-persistent HTTP if the browser can open 3 parallel connections at the same time is approximately $4RTT + 0.7012$ sec. ☐

☒ True

☐ False

☒ Will be reviewed

44 2-point

Consider distributing a file of F bits to N peers using a client-server architecture. Denote the upload rate of the server's access link by u_s , and d_{min} denotes the download rate of the peer with the lowest download rate. Assume a fluid model where the server can simultaneously transmit to multiple peers, transmitting to each peer at different rates, as long as the combined rate does not exceed u_s . Suppose that $u_s/N \leq d_{min}$, then the distribution scheme that has a distribution time of NF/u_s is..... ☐

☒ a distribution scheme in which the server sends the file to each client, in parallel, at a rate of u_s/N

☐ a distribution scheme in which the server sends the file to each client, in parallel, at a rate of d_{min}

☐ a distribution scheme in which the server sends the file to each client, in serial, at a rate of u_s/N

☐ a distribution scheme in which the server sends the file to each client, in serial, at a rate of d_{min}

☐ None of the above

☒ Will be reviewed

45 0.5-point

In CSMA, if the adapter senses that the channel is idle, it starts to transmit the frame. If the adapter senses that the channel is busy, it waits until it senses no signal energy and then starts to transmit the frame. While transmitting, the adapter monitors for the presence of signal energy coming from other adapters that use the channel. ☐

☒ True

☐ False

☒ Will be reviewed

46 1-point

Ethernet technology provides connectionless service to the network layer. Also, it provides an unreliable service to the network layer in which when an adapter “B” receives a frame from “A”, it runs the CRC check, but it doesn't send an acknowledgment or negative acknowledgment. When a frame fails the CRC check, adapter “B” discards the frame. ☐

☒ True

☐ False

📧 Will be reviewed

47 0.5-point

Suppose two IPv6 nodes want to communicate using IPv6 datagrams in which the receiver node needs to receive unchanged IPv6 datagram. But the two nodes are connected to each other by IPv4 routers. Then the preferred method in this case will be

☐ using dual-stack approach

☒ using Tunneling

☐ using NAT

☐ None of the above

📧 Will be reviewed

48 0.5-point

In case of sending the frame broadcast, each adaptor that receives the frame will check to see whether the destination MAC address in the frame matches its own MAC address. If it doesn't match, the adapter discards the frame without passing the network-layer datagram up the protocol stack.

☐ True

☒ False

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